The Physics of an Incomplete Crossing: A Single Conservation Law under Restricted Access

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Abstract

We take one physical premise — a globally conserved Noetherian flow, $\partial_I J^I = 0$, observed under restricted access — and ask what structure it forces when no part of the description is given in advance: no spacetime, no metric, no gauge group, no field content, no time. The premise is shown to be inevitable rather than chosen: a finite observer reading a conserved flow can read it only through a map of finite rank, which cannot be injective, so the crossing from what is conserved to what is observed is necessarily incomplete. From this single fact a chain of consequences follows, each derived here in full. Global conservation under restricted access requires a compact internal phase, whose carrier must be double-valued under closed transport and hence a self-conjugate (Majorana) field. The incompleteness of the crossing is a non-trivial kernel; what the kernel cannot return appears at the boundary as a non-linear tension. The carrier's self-conjugacy, together with the primitive second-order closure of its internal scale, forces a Fibonacci law on the norms of its modes, and the condition that the channel count equal its own closure capacity selects the closure capacity $W = 5$ uniquely — the coincidence of this capacity with the metric (SVD) rank of the projection being a separate, downstream result. The minimal domain on which the projected observables are non-degenerate is $\mathbb{R}^4 \times S^1(\tau)$, forced as a representational structure and not posited as a five-dimensional spacetime; the only observables are bilinears in the carrier; and time is identified, not assumed, as the irreversible ordering of the coherence that does not cross. This is the whole content of the present paper. The quantitative invariants of the theory — the fine-structure constant, the gauge algebra, the electroweak mixing angle — are *not* derived here; they are downstream consequences of this structure, established in companion papers, and are collected in a dependency map so that what this axiom forces is never confused with what the chain later proves.

Keywords: Noetherian conservation; restricted accessibility; non-invertible projection; Majorana carrier; compact internal phase; rank selection; emergence of time; axiomatic foundations.

1 Introduction: why the crossing is incomplete

The deepest difficulty in founding physics on something other than physics is that the founder keeps smuggling in what was meant to be derived. One sets out from "first

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principles" and finds, on inspection, that a spacetime was assumed, or a metric, or a symmetry group, or a notion of evolution in time. The aim of this paper is to begin from a single premise that contains none of these, and to show that the structure of observable physics is fixed by that premise once one condition is imposed: that the flow be observed by something finite.

The premise is a conservation law. Let there be a globally conserved Noetherian flow,

$$\partial_I J^I = 0, \tag{1}$$

where the index I ranges over a domain left deliberately unspecified — not assumed to be spacetime, not carrying a metric, not equipped with an equation of motion. Equation (1) is the weakest statement under which a conserved quantity retains its identity as it is redistributed. By itself it says nothing observable. It acquires content the moment access to the flow is restricted: when only a part of the invariants of J^I is operationally available to a reduced observer.

That this restriction is not an extra assumption but a necessity is the first thing to establish, and it is what makes (1) a legitimate starting point rather than a smuggled structure. A physical observer is finite: it registers a finite number of independent quantities. Whatever map carries the conserved flow to what the observer can hold therefore has finite rank. If the flow itself has more independent structure than the observer can register — as any field on an extended domain does — then the map from flow to observation cannot be injective: distinct states of the flow must collapse to the same observation. The crossing from what is conserved to what is observed is, of necessity, incomplete. Restricted access is not imposed on (1); it is what observation of (1) means.

This paper derives, in full and from (1) together with the finiteness of observation, the structural skeleton of the observable world:

$$\begin{aligned} \partial_I J^I = 0 \ \xrightarrow{\text{restricted access}}\ \chi \in S^1 \ \Rightarrow\ \ker \Pi \neq 0 \ \Rightarrow\ \Xi = C\bar{\Xi} \\ \Rightarrow\ \mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau) \ \Rightarrow\ \mathcal{O} = \bar{\Xi} M \Xi \ \Rightarrow\ \text{time.} \end{aligned} \tag{2}$$

Every arrow in (2) is proved here, with hypotheses stated and conclusions bounded. What is *not* proved here — the fine-structure constant, the gauge group, the mixing angle, and the rest of the quantitative content of the theory — is downstream of this skeleton, derived in the companion papers of the series and listed in the dependency map of Section 9. The discipline of the present paper is to force the question, not to answer all of it: to show that a conserved flow under restricted access *cannot but* produce a non-invertible projection with a kernel, a self-conjugate carrier, a four-plus-one representational domain, bilinear observables, and an arrow of time, and to leave the numbers to the chain.

The order of exposition is structural, then ontological: each consequence is forced by a stated necessity before its meaning is drawn out, so that the interpretation rests on the derivation and never precedes it.

2 The axiom

We state the single premise and fix the smallest vocabulary it requires. Nothing in this section presupposes the conclusions of later sections; the projection, its kernel, and the domain are introduced as the structures the axiom forces, and are constructed where they first become necessary.

Axiom 1 (Noetherian conservation under restricted access)**.** There exists a globally conserved Noetherian flow,

$$\partial_I J^I = 0, \tag{3}$$

observed under restricted access: not all invariants of J^I are operationally available to a reduced observer. No metric, dynamical law, dimensionality, or field content is assumed; the domain of the index I is to be determined, not posited.

The two clauses of the axiom pull in opposite directions, and physics is the resolution of the tension between them. Conservation is global: the quantity $Q[\Sigma] = \int_\Sigma J^I \, d\Sigma_I$ is invariant under deformation of the cut Σ. Access is restricted: a reduced observer cannot read all of J^I. The conserved whole must therefore split into a part that is read and a part that is not, in such a way that the whole is still conserved. Everything below is the working-out of that split.

Definition 2.1 (Bulk, observation, kernel)**.** The *bulk* is the separable Hilbert space $\mathcal{H}_{\text{bulk}}$ of configurations of the conserved flow. *Observation* is a bounded linear map $\Pi : \mathcal{H}_{\text{bulk}} \to \mathcal{H}_{\text{obs}}$ onto the space of data a reduced observer can hold, with the observed field $\Psi = \Pi\,\Xi$ for a bulk carrier Ξ. The *kernel* $\ker \Pi = \{\Xi : \Pi\Xi = 0\}$ is the set of bulk configurations producing no observation. Two configurations are observationally identical iff their difference lies in $\ker \Pi$, so the observable world is the quotient $\mathcal{H}_{\text{bulk}} / \ker \Pi$.

These are the only primitive notions. That Π exists, has finite rank, and has a non-trivial kernel is not assumed in Definition 2.1; the axiom forces it in Section 4, once the carrier of the flow has been identified.

3 The compact phase and its carrier

Before there can be a projection there must be something to project. The axiom constrains that something tightly: it must be able to carry a globally conserved flow whose phase is, in part, inaccessible. We derive first the geometry of the inaccessible part — a compact internal phase — and then the algebraic nature of its carrier — a self-conjugate spinor. Both are forced; neither is posited.

3.1 The inaccessible phase is compact

Theorem 3.1 (A compact internal phase is necessary)**.** *Under Axiom 1, together with the* finite-return requirement *— that the inaccessible carrier remain globally identifiable after transport, so that conservation can be checked across successive cuts — the inaccessible part of the conserved flow is carried by a compact internal phase $\chi \in S^1$. The non-compact line $\mathbb{R}$ is excluded; no higher-dimensional compact group is required.*

Proof. The conserved flow has a globally transportable mode — the component invariant under local redistribution, which carries the conserved quantity around any closed cut. Restricted access removes this mode from direct observation. Conservation under restricted access does not by itself forbid an unbounded carrier: a flow on $\mathbb{R}$ can be conserved. What the framework requires is the finite-return condition stated in the hypothesis — that the inaccessible mode be globally identifiable after transport, so that what is not read at one cut can be matched to what is read at the next. This requires a non-trivial closed transport: a recurrence with a well-defined return map. A non-compact one-parameter group $\mathbb{R}$ has no intrinsic return — transport along $\mathbb{R}$ never re-identifies any two configurations, so the inaccessible mode could not be matched across cuts and the consistency check would fail. The minimal connected one-dimensional group admitting closed orbits, finite holonomy, and a return map is the circle S^1. A higher-dimensional compact group would introduce further independent internal directions not required by a single conserved flow, violating minimality. Hence the inaccessible phase is exactly $\chi \in S^1$. □

3.2 The carrier is self-conjugate

The compact phase must be carried by a field. The axiom and the compactness just derived constrain that field to a single algebraic type. We establish the two properties that fix it: double-valuedness under closed transport, and self-conjugacy.

Theorem 3.2 (The carrier is double-valued)**.** *Under Axiom 1, the carrier Ξ of the conserved flow on S^1 must change sign under transport around the compact phase,*

$$\Xi(\chi + 2\pi) = -\,\Xi(\chi). \tag{4}$$

A single-valued carrier produces no non-trivial observable structure.

Proof. Let $\hat{H}$ be the holonomy operator of one transport around S^1. A single-valued carrier has $\hat{H} = +1$: every closed transport returns the carrier unchanged, and the only invariant the observation can extract is the identity, which carries no positional information around the phase cycle — no non-trivial boundary algebra is generated. The minimal non-trivial alternative is the sign reversal $\hat{H} = -1$, equation (4): the carrier returns to itself only after two transits, and its holonomy satisfies the minimal polynomial $z + 1 = 0$ over $\mathbb{Z}$, generating a two-dimensional boundary algebra. Any holonomy of higher order would introduce an additional independent phase sector, exceeding what a single conserved flow supplies. Double-valuedness is therefore the minimal — and so the forced — non-trivial holonomy. □

Theorem 3.3 (The carrier is Majorana)**.** *Under Axiom 1, the double-valued carrier must be self-conjugate,*

$$\Xi = C\bar{\Xi}, \tag{5}$$

with C the charge-conjugation operator. The Majorana carrier is the minimal admissible one.

Proof. The carrier must satisfy four conditions simultaneously: it carries a single globally conserved flow (global neutrality, not two independent charge sectors); it has the non-trivial holonomy (4); the split between what crosses the projection and what remains is internal to the field, not imposed from outside; and it presupposes no metric, since the metric is itself to be produced by the projection. We eliminate the alternatives. A *scalar* has no spinorial orientation and cannot supply the two chiral channels that the double-valued phase requires; its projection reads only an amplitude. A *vector* carries a spacetime index and requires a metric to contract it — unavailable at this level, and circular to assume. A *generic Dirac spinor* ($\psi \neq C\bar{\psi}$) contains independent particle and conjugate sectors that transform separately; under the single-flow axiom these are redundant degrees of freedom carrying more conserved channels than one flow supplies, unless the carrier is constrained by self-conjugacy. A *Weyl spinor* supplies one chirality only and leaves its conjugate sector independent and incomplete on the compact fibre. The minimal admissible carrier is therefore the one that contains its own conjugate internally: the self-conjugate Majorana spinor (5), which carries a single neutral flow, holds the non-trivial holonomy in its spinorial structure, splits internally into what crosses and what remains, and needs no metric. It is the unique carrier meeting all four conditions. □

The conserved flow can now be written in the language of the carrier as the Noether bilinear $J^I = \bar{\Xi}\Gamma^I\Xi$ with $\partial_I J^I = 0$; the axiom has acquired a body without acquiring a new assumption. The carrier decomposes, under the action of the compact phase, into two chiral channels $\Xi \mapsto (\xi, \eta)$ — a longitudinal ξ and a transverse η — whose roles are fixed only later by the projection.

4 The projection and its kernel

The carrier exists; the compact phase exists; the conservation law has a body. Nothing has yet been observed. We now show that observation forces a non-invertible projection with a non-trivial kernel, and we construct the kernel as the locus of the coherence that does not cross.

Theorem 4.1 (The projection is non-invertible)**.** *Under Axiom 1, observation of the carrier is a bounded linear map* $\Pi : \mathcal{H}_{\text{bulk}} \to \mathcal{H}_{\text{obs}}$ *that is not injective:* $\ker \Pi \neq \{0\}$.

Proof. Observation is finite (Section 1): the reduced observer registers a finite number of independent quantities, so Π has finite rank, $\dim \operatorname{Im}(\Pi) < \infty$. The bulk carrier on the compact-phase-extended domain has infinitely many independent configurations, $\dim \mathcal{H}_{\text{bulk}} = \infty$. A map from an infinite-dimensional space to a finite-rank image cannot be injective; equivalently, restricted access (Axiom 1) states directly that the observer cannot reconstruct all bulk invariants, which an injective Π would permit. Hence Π is non-invertible and $\ker \Pi \neq \{0\}$. □

Remark 4.2 (The kernel is the body of the inaccessible). The kernel is not a list of states the observer happens to miss; it is the equivalence relation that observation imposes on the bulk (Definition 2.1). What does not cross the projection is not destroyed — conservation forbids that — but retained in $\ker \Pi$ and returned, in part, to the boundary. The deficit between what the bulk conserves and what the boundary reads is exactly the kernel; it is the structural origin of everything the boundary cannot derive from within itself. A companion analysis shows, with no assumption on field content, that the kernel of any finite-rank non-invertible observation is infinite-dimensional and that its absolute scale lies in the kernel of the adjoint: so the boundary fixes the shape of the world but not its unit, which must be set by one external anchor [1].

What the kernel retains presses back on the boundary. The component of the carrier that fails to cross cannot vanish without breaking conservation; it returns as a non-linear tension on the observed field. Writing $\Psi = \Pi\Xi$ for the boundary field, this return is a boundary potential $V'(|\Psi|)$, the boundary's record of the coherence held in the kernel. The breathing exchange between boundary and kernel, in which the boundary yields coherence to the kernel and the kernel returns it, is the dynamical content that Section 8 identifies as time.

5 The closure capacity

The projection is non-invertible, but the integer governing how many independent channels its boundary can sustain is not yet fixed. We show that the self-conjugacy of the carrier, *together with* the primitive second-order closure of its internal scale, forces a Fibonacci law on the norms of its modes, and that the condition that the channel count equal its own closure capacity selects the value five. The integer derived here is the *closure capacity* W of the Majorana–Fibonacci carrier; that the metric (SVD) rank of the projection equals this capacity, $r_{\mathrm{SVD}} = W = 5$, is a separate result, proved downstream from the singular spectrum in ProjectionSVD [2] and recorded in Table 1. The present paper derives W; it does not derive the SVD rank.

Theorem 5.1 (Majorana with second-order closure yields the Fibonacci law). *Assume the Majorana condition* (5) and *the primitive second-order closure of the internal scale (hypothesis (C): the carrier's internal recurrence is of minimal second order, with the golden self-similarity fixed in Theorem 6.1). Then the squared norms of the carrier modes, ordered by internal scale, satisfy the Fibonacci recurrence*

$$\|\Xi\|_{n+1}^2 = \|\Xi\|_n^2 + \|\Xi\|_{n-1}^2. \tag{6}$$

Proof. Expand the carrier in modes $\{\Xi_n\}$ ordered by scale. Self-conjugacy $\Xi = C\bar{\Xi}$ requires each mode Ξ_n to sit in a space that also contains its charge conjugate $C\overline{\Xi_n}$; the two are orthogonal in the Clifford inner product, lying in opposite chiral sectors. The mode at the next level is their direct sum,

$$\Xi_{n+1} = \Xi_n \oplus C\overline{\Xi_n}.$$

Charge conjugation is an isometry, so $\|C\overline{\Xi_n}\| = \|\Xi_n\|$. This much follows from self-conjugacy alone, and gives only $\|\Xi_{n+1}\|^2 = 2\|\Xi_n\|^2$. To obtain the Fibonacci relation rather than mere doubling, hypothesis (C) is needed: the primitive second-order closure fixes the golden self-similarity of the internal scale (Theorem 6.1), under which the conjugate norm at level n equals the norm at level $n-1$, $\|C\overline{\Xi_n}\|^2 = \|\Xi_{n-1}\|^2$. Taking norms then gives $\|\Xi_{n+1}\|^2 = \|\Xi_n\|^2 + \|\Xi_{n-1}\|^2$, which is (6). The recurrence follows from Majorana self-conjugacy *and* second-order closure together; self-conjugacy alone yields doubling, not the Fibonacci law. □

The n-th mode of (6) contributes F_n linearly independent channels, where F_n is the n-th Fibonacci number. A carrier supporting w independent channels closes — its growth in independent directions matches its capacity — exactly when the channel count and the Fibonacci index coincide.

Theorem 5.2 (The closure capacity is five). *The closure condition $F_w = w$, that the Majorana–Fibonacci channel count equal its own closure capacity, has a unique non-trivial solution,*

$$W = 5. \tag{7}$$

Proof. Inspect the sequence: $F_1 = 1 = 1$, $F_2 = 1 \neq 2$, $F_3 = 2 \neq 3$, $F_4 = 3 \neq 4$, $F_5 = 5 = 5$, and for $w > 5$ the Fibonacci growth, exponential in $\varphi = (1+\sqrt{5})/2$, strictly exceeds the linear count, $F_w > w$. The solutions are $w = 1$ (trivial: one channel, no boundary algebra) and $w = 5$. Hence the unique non-trivial closure capacity is $W = 5$. It is moreover prime, which rigidifies the selection: 5 is the unique prime fixed point of the Fibonacci map in this range, so the closure admits no proper sub-factorisation into independent channels. □

Remark 5.3 (Three logically separate facts). Three statements at different logical depths should not be conflated. The non-invertibility $\ker \Pi \neq \{0\}$ (Theorem 4.1) is pre-physical: it holds for any finite-rank observation of an infinite bulk. The closure capacity $W = 5$ (Theorem 5.2) is conditional on the Majorana–Fibonacci structure together with second-order closure: remove the carrier's self-conjugacy or the closure hypothesis and W is unconstrained. The metric (SVD) rank $r_{\mathrm{SVD}} = 5$ is a third statement, established independently from the singular spectrum of the projection in a companion paper [2], which proves $r_{\mathrm{SVD}} = W$. The present paper derives the closure capacity W, not the SVD rank.

6 The representational domain

We have a carrier, a projection, a kernel, and a rank. The domain on which all of this lives has not been posited — the axiom left the index I undetermined — and we now show it is fixed. The requirement that the projected bilinear observables be non-degenerate selects a four-dimensional non-compact factor; the compact phase supplies one further factor; and the separation between kernel and boundary forbids any twisting between them. The domain is $\mathbb{R}^4 \times S^1(\tau)$, a representational structure and not a five-dimensional spacetime.

Theorem 6.1 (The domain and its scale are necessary)**.** *Under Axiom 1, the minimal domain supporting non-degenerate projected bilinear observables is*

$$\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau), \tag{8}$$

and the second-order closure of the carrier fixes the spectral separation $\sqrt{5}$, hence the golden internal connection

$$\Gamma_\tau = \frac{\varphi}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{9}$$

The boundary carries a discrete conformal structure parametrised by φ, not a metric.

Proof. Four extended directions. The projection must act on a domain supporting all projected bilinear observables without a pre-assumed metric. The requirement is not the bare existence of a real spinor representation — real Clifford algebras admit representations across dimensions by the mod-8 periodicity — but the existence of a domain on which the *full non-degenerate real bilinear algebra* demanded by the projected carrier can be realised: the double-valued carrier of Theorem 3.2, its residual non-Abelian structure, and a non-degenerate set of real bilinears, all simultaneously and compatibly with the signature $(1,3)$ that the boundary later inherits. Four extended directions are the minimal representational threshold meeting this requirement; below four, the bilinear algebra degenerates and the carrier cannot be observed without collapse. The domain cannot be compact: compactness would enforce global identifications among projected observables, contradicting restricted access. Hence $\mathbb{R}^4$ is the smallest non-compact representational factor available.

One compact direction. The compact phase $\chi \in S^1$ (Theorem 3.1) requires exactly one compact internal direction along which global transport closes. A second compact direction would introduce an independent conserved channel, violating the single-flow clause of the axiom. Hence $S^1(\tau)$ is the unique compact factor.

No twisting. A non-trivial fibration would correlate the accessibility of the internal phase with the projected observables, violating the separation between $\ker \Pi$ and the boundary readout. The product is therefore direct, $\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau)$.

The scale. The carrier's holonomy closes at second order. The conjugate pair $(\lambda, -\lambda^{-1})$ forced by the spinorial sign (4) has, under restricted access, only its inversion-invariant separation accessible. The minimal closure polynomial $\lambda^2 - K\lambda - 1 = 0$ with the self-consistency $K = 1$ gives roots $\lambda_\pm = (1 \pm \sqrt{5})/2$ and discriminant $(\lambda_+ - \lambda_-)^2 = 5$. The accessible spectral separation is thus $\sqrt{5}$, and the native rate of the carrier is the golden connection $\Gamma_\tau = \varphi/\sqrt{5}$ of (9). The boundary inherits the golden self-similarity as a discrete conformal structure, not as a metric: φ scales multiplicatively, and the boundary reads ratios, not lengths. □

Remark 6.2 ($\mathbb{R}^4$ is not a background). The four-dimensional factor is not a stage assumed in advance and then populated. It is the representational structure forced by the demand that the projected bilinears be non-degenerate: the minimal arena in which the carrier can be observed at all. A theory formulated intrinsically on $\mathbb{R}^4$ from the outset cannot derive the algebra of admissible observables, nor the dimensionless invariants, nor the bulk-to-boundary split; these require the projection with non-trivial kernel, which $\mathbb{R}^4$ alone does not contain. The domain is a consequence of observation, not its precondition.

7 The observables

The structure is now fixed: carrier, compact phase, projection, kernel, rank, and domain. It remains to say what can be observed on it. The answer is determined by the same constraints, and it is restrictive: the only admissible observables are bilinears in the carrier.

Theorem 7.1 (Observables are bilinear)**.** *Under Axiom 1, every admissible observable is a real bilinear invariant of the carrier,*

$$\mathcal{O} = \bar{\Xi}\, M\, \Xi, \tag{10}$$

where M ranges over the internal algebra. No linear functional of the carrier survives the projection.

Proof. A linear functional of Ξ depends on the inaccessible phase χ: under $\chi \to \chi + 2\pi$ the carrier changes sign (4), so any linear quantity reverses and cannot be a well-defined observable on the quotient $\mathcal{H}_{\text{bulk}}/\ker \Pi$. A bilinear $\bar{\Xi} M \Xi$ is invariant under $\chi \to \chi + 2\pi$, since the two sign reversals cancel, and is therefore phase-blind:

it survives the projection. The self-conjugacy (5) makes the bilinears real. These are the only observables compatible with restricted access: they encode the carrier only through combinations that the projection does not annihilate. Any admissible observable is a real combination of such bilinears. $\square$

The bilinear $\mathcal{O} = \bar{\Xi} M \Xi$ is the general observable; the conserved Noether current $J^I = \bar{\Xi}\Gamma^I\Xi$ is the special case $M = \Gamma^I$. The two chiral channels (ξ, η) of the carrier enter observables only through their invariant combinations; in particular the boundary reads the product $\xi\eta$ as a single scalar, the invariant whose value is fixed downstream by the geometry of the projection (Section 9). At the level of the present paper, ξ and η are the longitudinal and transverse readings of one self-conjugate field, and the electromagnetic and gravitational sectors are recovered downstream as the two channel-readouts P_η and P_ξ of the boundary field $\Psi = \Pi\Xi$ — not as scalar limits of ξ or η, which the invariance of $\xi\eta$ forbids.

8 Time

One element of (2) remains: time. It is not among the inputs — the axiom has no evolution parameter — and we show it is forced as the ordering of the coherence that does not cross. This is the ontological terminus of the derivation: the structure cannot be described statically, and the minimal ordering it requires is what we call time.

Theorem 8.1 (Time is the ordering of irreversible loss). *Under Axiom 1, let $a(R) := \dim \mathcal{C}_{\mathrm{coh}}(R)$ denote the* accessible coherence *of a structural recurrence R — the dimension of the coherent sector still readable at the boundary in R. (In concrete realisations $a(R)$ may instead be represented by the accessible charge $Q_{\mathrm{acc}}(R)$ or by $|P_{\mathrm{coh}}J(R)|^2$; the argument needs only that a be a real-valued functional that strictly decreases under loss.) The boundary–kernel exchange is irreversible, a is strictly monotone along it, and the relation*

$$R_i \prec R_j \quad \Longleftrightarrow \quad a(R_j) < a(R_i) \tag{11}$$

is a partial order on the structural recurrences of the projected flow. Time is identified with this order; it is not a coordinate.

Proof. Each structural recurrence is one complete instance of the balance between the accessible boundary sector and the inaccessible kernel. The projected flow does not conserve locally: $\partial_I (\Pi J)^I \neq 0$, the deficit being carried by the kernel through the global balance. The projection Π is non-invertible (Theorem 4.1), so the coherence transferred to the kernel at each recurrence cannot be projected back; hence the accessible coherence strictly decreases along the exchange, $a(R_j) < a(R_i)$ whenever R_j is reached from R_i, and the functional a is well defined and real-valued on recurrences. With $\prec$ defined by (11), antisymmetry is immediate: $R_i \prec R_j$ and $R_j \prec R_i$ would require $a(R_j) < a(R_i)$ and $a(R_i) < a(R_j)$ simultaneously, impossible for real numbers. Transitivity is the transitivity of $<$ on a-values: $a(R_k) < a(R_j) < a(R_i)$ gives $a(R_k) < a(R_i)$. Non-totality reflects that recurrences with equal a but different internal organisation are incomparable, since (11) relates only strictly ordered a-values. The relation $\prec$ is therefore a partial order, forced by the strict monotonicity of a under irreversible loss and by nothing else. A purely cyclic identification of recurrences would erase the irreversibility (it would set a constant) and is excluded; a total order would impose a global comparability among equal-a recurrences that restricted access denies. The partial order induced by the decay of a is the unique admissible ordering, and it is what the theory calls time. □

Remark 8.2 (The static description is impossible). The force of Theorem 8.1 is that the structure derived in this paper cannot be held still. A conserved flow under restricted access loses coherence to the kernel at each recurrence and cannot recover it; there is no description in which the recurrences are simultaneous and equivalent. Ordering is not added to the structure; it is what remains when nothing else can be added. Matter, in the same register, is what survives the loss: the bilinear invariants of the carrier that persist as coherence erodes. What orders the loss is time; what persists through it is matter. Neither is a primitive; both are readings of a single conserved flow crossing incompletely into observation.

9 Downstream consequences and dependency map

The present paper derives the structural skeleton of the observable world and nothing more. Everything quantitative — the value of the fine-structure constant, the gauge algebra, the electroweak mixing angle, the matter spectrum, the cosmological readout — is a downstream consequence of this skeleton, established in the companion papers of the series. We list these consequences explicitly, with the paper in which each is derived, so that what *this* axiom forces is never confused with what the chain later proves. None of the entries in Table 1 is claimed as a result of the present paper; each is a theorem of the cited work, resting on the structure derived here.

The dependency is one-directional. The skeleton of Sections 2–8 is logically prior; the quantitative invariants are its consequences, read off at specific levels of the projection by the methods of the companion papers. One external dimensional input is required to close the dimensional structure — a single metrological anchor, identified in the series with the Rydberg constant R_∞ [6] — in accordance with the general result that a finite-rank projection fixes every dimensionless relation but not the absolute scale [1]. The axiom forces the structure; the structure forces the question; the chain answers it.

10 Conclusion: the single statement

One premise has carried the whole construction: a conserved Noetherian flow, $\partial_I J^I = 0$, observed under restricted access — with the restriction shown to be not an assumption but the meaning of finite observation. From it, with no spacetime, metric, gauge group, field content, or time given in advance, the following are forced, each derived in full above. A compact internal phase carries the inaccessible part of the flow. Its carrier is double-valued under closed transport and hence a self-conjugate Majorana field. Finite observation is a non-invertible projection whose kernel holds the coherence that does not cross. The carrier's self-conjugacy, with the primitive second-order closure of its internal scale, forces a Fibonacci law on its mode norms, and the demand that the channel count equal its own closure capacity selects the closure

Downstream structure	Skeleton element it rests on	Derived in
Metric rank equals the closure capacity	closure capacity W (§5)	ProjectionSVD [2]
Boundary metric and rational filter	metric rank, Pincherle scale	ProjectionSVD [2]
Projective pole of the boundary filter	filter / Pincherle structure	ProjectionSVD [2]
Gauge factorisation of the boundary algebra	Majorana boundary roles (§5)	GaugeAlgebra [3]
Mixing angle as a ratio of mode counts	boundary mode count	GaugeAlgebra [3]
Phasorial–spinorial kernel: Hardy split, Heegner index	kernel structure (§4)	KernelSpinor [4]
Physical coupling as the boundary fixed point	projective pole + breathing dynamics	Alpha [7]
Born rule, measurement, breathing on Σ	bilinear observables (§7)	BoundaryAlgebra [5]
Atomic length scale and its metrological anchor	pseudoinverse of Π	BohrRydberg [6]
Scale-free residue in the incoherent limit	non-invertible projection (§4)	StructuralConstraints [1]

Table 1: Downstream consequences of the axiomatic skeleton, named by their *structure*, not by their numerical value: this paper fixes none of the numbers. Each entry is a theorem of the cited companion paper; the middle column names the skeleton element — forced here — on which it rests.

capacity $W = 5$ uniquely (its coincidence with the metric rank being proved downstream). The minimal domain on which the projected bilinears are non-degenerate is $\mathbb{R}^4 \times S^1(\tau)$, a representational structure rather than a spacetime, carrying a golden internal connection $\Gamma_\tau = \varphi/\sqrt{5}$. The only admissible observables are bilinears in the carrier. And time is the irreversible ordering of the coherence the projection cannot return.

This is the axiomatic core, and it is deliberately silent about numbers. The fine-structure constant, the gauge algebra, the mixing angle, and the rest are downstream of this skeleton (Section 9), proved in the companion papers and not here. The contribution of the present work is to make the question inevitable: given a conserved flow and restricted access, the observable world *must* be the boundary of a non-invertible projection with a non-trivial kernel, carried by a self-conjugate spinor, ordered by the loss that the projection forces. Stated once:

> *Given a conserved flow and restricted access, observable physics requires a non-invertible projection with a kernel; what crosses is the world, what does not is its memory, and the order of the loss is time.*

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Companion papers

The following are the companion papers of the series in which the downstream consequences of Table 1 are derived. The present paper depends on none of them; they depend on it.

1. P. Camelia, *What a Finite Observer Cannot Know: Structural Constraints on Observable Algebras from Non-Invertible Projections* (2026).
2. P. Camelia, *Projection-Induced SVD and Boundary Metric: A Rank-Five Fibonacci-Auto-similar Characterization* (2026).
3. P. Camelia, *Observable Gauge Algebra from Finite-Rank Majorana Projection* (2026).
4. P. Camelia, *The Kernel Spinor* $\Xi_{\mathrm{ker}}(163)$*: Domain Walls, the Hilbert–Hardy Filter, and the Ramanujan–Heegner Index* (2026).
5. P. Camelia, *The Boundary Algebra of a Non-Invertible Projection* (2026).
6. P. Camelia, *The First Dimensional Ruler: Bohr Radius and Rydberg Constant from the QGT Projection* (2026).
7. P. Camelia, *The Self-Consistent Fine-Structure Constant: Breathing Correction, Tensorial Structure, and the Fixed Point of the Observable Boundary* (2026).